Venkata Lakshmi Parimala Pasupuleti

AI Engineer · Production LLM Systems · RAG Pipelines · Agentic Workflows · LLM Evaluation · LangGraph · ChromaDB · HITL · Python · TypeScript · Healthcare AI · MCP

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PROFESSIONAL SUMMARY

AI Engineer with 8 years of production engineering experience and two shipped LLM-based systems built on the exact stack Machinify targets: LangGraph, ChromaDB, HITL escalation, production observability, and evals-before-prompts. AgentProbe is an open-source evaluation suite built on the philosophy that prompts and evals are code - 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge scoring, CI-integrated regression gate, and multi-provider LLM routing (Claude, GPT, Gemini, Ollama) for cost-aware optimization. LeaseI is an 8-agent LangGraph platform with multi-source RAG, streaming stateful workflows, deterministic guardrails on regulated outputs, and HITL escalation when model confidence falls below threshold - 93.6% precision across 1,240 compliance test cases, zero regressions shipped. Healthcare AI domain: built AI on FHIR, a HIPAA-compliant NLP system for medical records using BERT embeddings, LLMs, and RBAC-enforced data isolation. Production infrastructure depth from Entrata: 99.5% uptime, event-driven async processing (RabbitMQ), cost-controlled pipelines, and observability tooling - the unglamorous infrastructure that makes systems work at scale.

TECHNICAL SKILLS

AI Engineering: LLM Applications, RAG Pipelines, Agentic Workflows, LangGraph, LangChain, Multi-Agent Orchestration, LLM Evaluation (evals-first), LLM-as-Judge, CI-Integrated Regression Gating, Prompt Engineering, Function Calling, Structured Outputs, Context Management, ChromaDB, Vector Search, Embeddings, HITL, Hallucination Detection, Guardrails, Cost Controls, Observability, OpenAI API, Anthropic API, Groq, Ollama (open-weight), MCP, BERT, spaCy, NLP

Engineering & Production: Python (8yr), TypeScript, JavaScript, FastAPI, Flask, SQL, Docker, Microservices, REST APIs, CI/CD (GitHub Actions), AWS (S3, EC2, RDS, Glue), RabbitMQ, Redis, PostgreSQL, MySQL, Oracle, Bash, Linux, Git, Streaming Systems

Data & Healthcare: Healthcare AI (FHIR, HIPAA, RBAC), PHI Compliance, PyTorch, scikit-learn, NumPy, Pandas, PySpark, Feature Engineering, Statistical Modeling, Tableau, Agile/Scrum

EXPERIENCE

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built end-to-end LLM-based agentic system (LangGraph) for mission planning - RAG pipeline with multi-source retrieval, streaming stateful workflows, and deterministic guardrails enforcing compliance rules before delivery; CI/CD-integrated evals blocking regressions before every release.
- Implemented cost-aware multi-provider LLM routing; observability on retrieval quality, latency, and token usage; HITL escalation for edge cases requiring human review.

George Mason University

Aug 2024 - May 2026

Graduate Teaching Assistant & Research Associate

- Built Python and R ML pipelines (scikit-learn, PyTorch) on WHO and UK Road Safety datasets - $R^2 = 0.977$; end-to-end from data ingestion through feature engineering, model evaluation, and validation across 190+ countries.
- Mentored 600+ students in Python, SQL, and data science; designed dashboards improving research comprehension by 40%.

Entrata - PropTech / Affordable Housing SaaS

Sep 2018 - Aug 2024

Senior Software Engineer · Lehi, UT (Remote)

- **Owned production data pipelines at 99.5% uptime** - annual HUD income-limit import (federal source → 180,000 domain records) powering eligibility, rent, and compliance calculations for 100K+ housing units; validation, reconciliation, and rollback preparation at every release; zero compliance violations across 7 years.
- **Built the unglamorous infrastructure that keeps systems reliable** - RabbitMQ async processing with retry, idempotency, dead-letter guarantees; Redis caching; observability and monitoring tooling reducing triage time 40%; sub-10 bug count for 6+ consecutive months at scale.
- **Cost controls and performance engineering at production scale** - 200+ SQL/PostgreSQL/Oracle query optimizations (30% performance gain); AWS S3, EC2, RDS; Docker microservices redesign (50% scalability improvement, 30% deployment reduction); Python/Bash automation eliminating 80% of manual operational workload.
- 100% PHI compliance with Libsodium-encrypted pipelines; 300+ code reviews; mentored 5+ engineers on production engineering and AI system design standards.

Entrata India - Pune

Jan 2017 - Sep 2018

Associate Software Engineer → Trainee

- Built backend modules (PHP, Python, PostgreSQL); data pipeline for 1.6M HUD/Tax Credit records; AWS S3 migration resolving 90% of legacy storage constraints.

PROJECTS

AgentProbe: Open-Source LLM Evaluation Suite

Evals · Open-Source

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · OpenAI · Anthropic · Ollama | github.com/venkatalakshmiparimala/AgentProbe

Evals-before-prompts, in production: 15-endpoint FastAPI evaluation platform, 5-metric quality engine (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge auto-rater, CI-integrated regression gate, cost-aware multi-provider LLM routing (Claude/GPT/Gemini/Ollama). 1,240+ evaluation test cases - zero regressions shipped to production. Open-source contribution to AI tooling ecosystem.

Leasel: 8-Agent LangGraph Agentic AI Platform

Agentic · RAG · HITL

Python · LangGraph (8-agent) · FastAPI · TF-IDF · OCR · SQLite | github.com/venkatalakshmiparimala/Lease_complaince_rag

Production LLM-based system: 8-agent LangGraph stateful pipeline with streaming workflows, multi-source RAG, deterministic guardrails controlling all regulated outputs, and HITL escalation when model confidence falls below threshold (failure handling at the seam between LLM and production). 93.6% precision, 87.9% F1 across 1,240 compliance test cases; CI quality gate with full observability and telemetry.

AI on FHIR: Healthcare NLP Query System

Healthcare · NLP

Python · Flask · LLMs · FHIR · React · Docker · spaCy · BERT · Embeddings | github.com/venkatalakshmiparimala/AI_on_FHIR

Healthcare AI system combining BERT embeddings and LLM inference to translate natural-language queries into FHIR API calls - reduced clinical retrieval time ~60%; HIPAA-compliant RBAC with full data isolation. Healthcare domain credibility directly relevant to Machinify's payer-side intelligence platform.

EDUCATION

M.S. Data Analytics Engineering - George Mason University, Virginia — May 2026 · GPA 3.87

B.Tech Computer Science & Engineering - JNTU Anantapur — 2013-2017